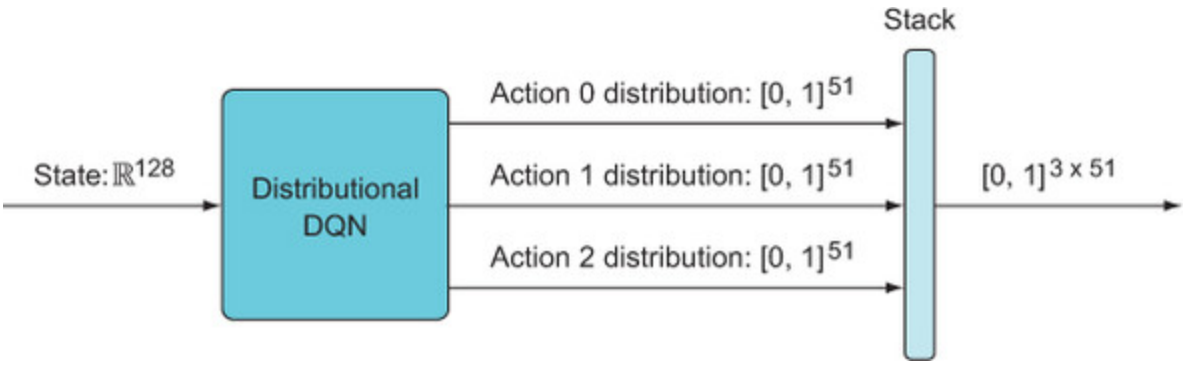


Listing 7.6. The Dist-DQN

```
def dist_dqn(x,theta,aspace=3):                          1
    dim0,dim1,dim2,dim3 = 128,100,25,51                  2
    t1 = dim0*dim1
    t2 = dim2*dim1
    theta1 = theta[0:t1].reshape(dim0,dim1)              3
    theta2 = theta[t1:t1 + t2].reshape(dim1,dim2)
    l1 = x @ theta1                                      4
    l1 = torch.selu(l1)
    l2 = l1 @ theta2                                     5
    l2 = torch.selu(l2)
    l3 = []
    for i in range(aspace):                              6
        step = dim2*dim3
        theta5_dim = t1 + t2 + i * step
        theta5 =
theta[theta5_dim:theta5_dim+step].reshape(dim2,dim3)
        l3_ = l2 @ theta5                                7
        l3.append(l3_)
    l3 = torch.stack(l3,dim=1)                           8
    l3 = torch.nn.functional.softmax(l3,dim=2)
    return l3.squeeze()
```

- ***1*** x is the 128-element vector state, theta is the parameter vector, and aspace is the size of the action space.
- ***2*** Defines the layer dimensions so we can unpack theta into appropriately sized matrices
- ***3*** Unpacks the first portion of theta into the first layer matrix
- ***4*** The dimensions of this computation are B × 128 × 128 × 100 = B × 100, where B is batch size.
- ***5*** The dimensions of this computation are B × 100 × 100 × 25 = B × 25.
- ***6*** Loops through each action to generate each action-value distribution
- ***7*** The dimensions of this computation are B × 25 × 25 × 51 = B × 51.